



AI INNOVATION HACKATHON 2026

Trustworthy AI

STANDARDIZED PROBLEM STATEMENTS

Women in AI, Research, Innovation & Entrepreneurship Club

Research • Innovation • Entrepreneurship

4 Challenges • Online • AI Innovation Hackathon 2026

Each challenge follows the same structured format for clarity, consistency and ease of review.

CHALLENGE PORTFOLIO

NO.	CHALLENGE	DIFFICULTY
1	TruthLens: Detecting Hallucinations and Verifying Citations in AI Chatbots	Intermediate to Advanced
2	FairDecide: Auditing and Reducing Bias in Loan and Hiring Decisions	Beginner to Intermediate
3	KnowWhenToStop: Uncertainty-Aware Chest X-Ray Triage Under Distribution Shift	Advanced (Intermediate with MedMNIST/CIFAR-10-C version)
4	GuardRail: Defending LLM Applications Against Jailbreaks and Prompt Injection	Intermediate



CHALLENGE 01

TruthLens: Detecting Hallucinations and Verifying Citations in AI Chatbots

DOMAIN

Hallucination & Citation Verification

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

Large Language Models are now used to answer medical, legal and educational questions, yet they regularly produce statements that sound confident but are factually wrong, and they sometimes invent references that do not exist. Retrieval-Augmented Generation (RAG) reduces this problem but does not remove it, because a model can still ignore, misread or contradict the documents it was given. Today, a user reading a chatbot answer has no practical way to know which sentences are backed by a reliable source and which are fabricated. This lack of verifiability is one of the main barriers to trusting AI in high-stakes settings. Build an AI system that, given a question, a set of source documents and a chatbot answer, automatically checks the answer claim by claim and tells the user how far it can be trusted.

EXPECTED SOLUTION

Deliver a prototype that breaks the answer into factual claims; labels each claim Supported, Contradicted or Not Enough Evidence and points to the exact source sentence used as evidence; produces an overall trust score/readable output; and reports performance on a held-out test set including partially correct answers.

SUGGESTED DATASETS / RESOURCES

TruthfulQA, HaluEval, FEVER, SciFact; Wikipedia dump; PubMedQA; optional Urdu/Roman Urdu test set.

AI / ML COMPONENT

Natural Language Inference (DeBERTa/RoBERTa); claim extraction; dense retrieval with embeddings + FAISS; LLM-as-judge; confidence calibration.

WHAT MAKES THE CHALLENGE INTERESTING?

A real, unsolved industry problem with a visually impressive demo; the optional Urdu angle gives local relevance.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter question + source documents + chatbot answer claims highlighted Supported/Contradicted/Not Enough Evidence evidence sentence trust score.

TECHNICAL REQUIREMENTS

Python; document parsing; retrieval; claim extraction; NLI; evaluation; confidence calibration; user-facing visualization.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Show claim-level labels and supporting evidence.
- Report held-out evaluation results including partially correct answers.

OPTIONAL ADVANCED FEATURES

- Add multilingual/Urdu evaluation.
- Compare retrieval/NLI pipelines.
- Add confidence calibration and abstention.
- Evaluate robustness to contradictory or incomplete sources.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Teams should use lawful/permitted datasets, protect sensitive information, document limitations and evaluate false positives/false negatives and robustness.

INNOVATION ANGLE

Combines claim-level verification, evidence retrieval and trust communication rather than only scoring an answer.

SUCCESS / EVALUATION METRICS

Claim classification accuracy/F1, evidence retrieval quality, calibration, false-support/false-contradiction rates and robustness.



CHALLENGE 02

FairDecide: Auditing and Reducing Bias in Loan and Hiring Decisions

DOMAIN

Fairness Auditing & Mitigation

DIFFICULTY

Beginner to Intermediate

PROBLEM STATEMENT

Banks and employers increasingly use machine learning models to approve loans and shortlist candidates. These models learn from historical data, which often reflects past discrimination based on gender, age, ethnicity or similar attributes. As a result, a model can achieve high overall accuracy while consistently rejecting qualified people from certain groups, and the organisation using it may never notice. Simply removing the sensitive attribute from the data is not enough, because other features (such as location or occupation) can act as proxies. Fairness also usually comes at some cost to accuracy, and decision makers need to understand this trade-off clearly.

EXPECTED SOLUTION

Build a toolkit/dashboard that audits a trained decision model for unfairness, applies at least one mitigation method, and explains individual decisions. Measure standard fairness metrics; show before/after results; visualize fairness-versus-accuracy trade-offs; and provide SHAP or counterfactual explanations.

SUGGESTED DATASETS / RESOURCES

Adult Income (UCI), German Credit, Folktables; COMPAS; Bank Marketing; Fairlearn; AIF360; SHAP.

AI / ML COMPONENT

Classification (XGBoost/neural networks); fairness-aware learning; explainable AI; counterfactual explanations.

WHAT MAKES THE CHALLENGE INTERESTING?

Shows that high overall accuracy is not automatically sufficient and makes trade-offs visible and measurable.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload/train model fairness metrics by group mitigation before/after comparison fairness/accuracy trade-off individual explanation.

TECHNICAL REQUIREMENTS

Python; tabular preprocessing; classification; fairness metrics; mitigation; XAI; dashboard; reproducible evaluation.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Measure fairness across sensitive groups.
- Apply at least one mitigation method and show before/after results.
- Explain individual decisions.

OPTIONAL ADVANCED FEATURES

- Compare multiple mitigation methods.
- Add proxy-feature analysis.
- Add threshold optimization and subgroup stress tests.
- Compare multiple XAI methods.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Teams should use lawful/permitted datasets, protect sensitive information, document limitations and evaluate false positives/false negatives and robustness.

INNOVATION ANGLE

Makes fairness auditing, mitigation and individual explanation part of one practical workflow.

SUCCESS / EVALUATION METRICS

Task performance, demographic parity/equal opportunity/equalized odds where appropriate, fairness-performance trade-off, explanation quality.



CHALLENGE 03

KnowWhenToStop: Uncertainty-Aware Chest X-Ray Triage Under Distribution Shift

DOMAIN

Uncertainty-Aware Medical AI

DIFFICULTY

Advanced (Intermediate with MedMNIST/CIFAR-10-C version)

PROBLEM STATEMENT

AI models for medical imaging often reach expert-level accuracy on the hospital data they were trained on. When the same model is used at a different hospital, with a different scanner or with a poor-quality image, its performance can drop sharply, and it usually still gives its wrong answer with high confidence. In healthcare, a silent and confident mistake is more dangerous than an honest 'I am not sure'. Clinicians need AI systems that recognise the limits of their own knowledge and hand difficult or unfamiliar cases over to a human expert.

EXPECTED SOLUTION

Build a chest X-ray classifier that predicts a diagnosis, estimates uncertainty, detects unfamiliar/corrupted images and defers such cases to a human. Train a classifier with calibrated confidence, detect OOD/corrupted images, implement abstention/defer, show coverage-vs-accuracy, and evaluate on a different source dataset.

SUGGESTED DATASETS / RESOURCES

CheXpert or NIH ChestX-ray14 for training; a different dataset for domain shift; CIFAR-10/CIFAR-10-C or MedMNIST as beginner-friendly alternatives; MC Dropout, Deep Ensembles, Temperature Scaling, Conformal Prediction.

AI / ML COMPONENT

CNN/Vision Transformer; uncertainty estimation; OOD detection; calibration; selective prediction; optional adversarial robustness testing.

WHAT MAKES THE CHALLENGE INTERESTING?

Teams are judged on how well the model knows its own limits, not only on accuracy; the organizers can offer easier and harder variants depending on compute.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload X-ray prediction + confidence OOD/quality check answer/defer decision coverage-vs-accuracy visualization.

TECHNICAL REQUIREMENTS

Python; medical/image classification; calibration; OOD detection; selective prediction; cross-dataset evaluation; visualization.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Output calibrated confidence.
- Detect OOD/corrupted inputs.
- Implement abstain/defer and report coverage-vs-accuracy.
- Evaluate on a different source where feasible.

OPTIONAL ADVANCED FEATURES

- Compare uncertainty methods.
- Add conformal prediction.
- Test robustness to controlled corruptions.
- Add clinician-oriented explanation of defer reasons.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Teams should use lawful/permitted datasets, protect sensitive information, document limitations and evaluate false positives/false negatives and robustness. For medical data, use only permitted datasets and clearly state that the prototype is not a clinical triage or diagnosis system.

INNOVATION ANGLE

Treats uncertainty and abstention as core system outputs rather than optimizing accuracy alone.

SUCCESS / EVALUATION METRICS

Accuracy/F1/AUROC where appropriate, calibration error, OOD detection performance, coverage-risk curve and cross-domain robustness.



CHALLENGE 04

GuardRail: Defending LLM Applications Against Jailbreaks and Prompt Injection

DOMAIN

LLM Security / Adversarial Robustness

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Companies are rapidly building applications on top of LLMs, such as customer support bots, email assistants and document readers. These applications can be attacked in two main ways: users can write jailbreak prompts that make the model ignore its safety rules, and attackers can hide malicious instructions inside web pages, emails or files that the model reads (prompt injection). Successful attacks can leak private data, trigger harmful actions or produce unsafe content. Most deployed applications have little or no defence, and simple keyword filters are easy to bypass. A good defence must also avoid blocking genuine users, otherwise the application becomes unusable.

EXPECTED SOLUTION

Build a guard layer in front of an LLM application that detects malicious or injected prompts and blocks or neutralizes them while allowing normal requests. Detect direct and indirect attacks; block/sanitize with a clear reason; report detection rate, false-positive rate and latency; and demonstrate a controlled red-team versus defence evaluation.

SUGGESTED DATASETS / RESOURCES

JailbreakBench, HarmBench, AdvBench; prompt-injection datasets on Hugging Face such as deepset/prompt-injections; BIPiA; benign prompts such as Alpaca/Dolly; small open models such as Llama, Mistral, Qwen.

AI / ML COMPONENT

Text classification (BERT/DeBERTa); embedding-based anomaly detection; LLM-as-a-guard; adversarial prompt generation/red-teaming; robustness evaluation.

WHAT MAKES THE CHALLENGE INTERESTING?

Highly relevant to LLM deployment; teams can play both attacker and defender and compare rule-based, ML-based and hybrid approaches.

SUGGESTED ON-SCREEN DEMONSTRATION

Send benign/malicious prompts and injected documents → guard decision → reason/sanitization → protected LLM response → red-team results dashboard.

TECHNICAL REQUIREMENTS

Python; safe sandboxed LLM testing; text classification/embeddings; attack dataset evaluation; latency measurement; logging and dashboard.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Detect jailbreak and prompt-injection attempts including indirect injections.
- Block or sanitize malicious input with a clear reason.
- Report detection rate, false-positive rate and response latency.

OPTIONAL ADVANCED FEATURES

- Compare rule-based, ML and hybrid defences.
- Generate controlled adversarial variants.
- Evaluate transfer to unseen attack patterns.
- Add document-level injection detection.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use only controlled test environments and public benchmark data. Do not attack third-party systems or expose real secrets. Keep red-team testing within the application and infrastructure authorized for the hackathon.

INNOVATION ANGLE

Evaluates security as an attacker-defender loop rather than a static keyword filter.

SUCCESS / EVALUATION METRICS

Attack detection rate/recall, false-positive rate on benign prompts, latency, robustness to unseen variants and successful attack rate.